

$$x^2 + y^2 - \left(\frac{y-3x}{c} \right)^2 = 1$$

$$x^2 \left(1 - \frac{36}{c^2} \right) + y^2 \left(1 - \frac{4}{c^2} \right) + \frac{24xy}{c^2} = 0$$

$$2 - \frac{40}{c^2} = 0 \Rightarrow c^2 = 20$$

(coefficient of x^2 + coeff. of y^2 = 0)

C02/1/022. If the line $lx + my = 4$ always touches the circle $x^2 + y^2 = 1$, then point (l, m) always lies on a circle whose area is :

- (A) π (B) 2π (C) 4π (D) 16π

Ans. D

Sol. $\frac{4}{\sqrt{\ell^2 + m^2}} = 1 \Rightarrow \ell^2 + m^2 = 16$

$\Rightarrow (l, m)$ lies on $x^2 + y^2 = 16$

$\Rightarrow$ Area of circle = 16π

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

C01/2/023. If $(\alpha, 0)$ lies inside the quadrilateral formed by lines

$$2x + 5y = 15,$$

$$5x - 4y = 21,$$

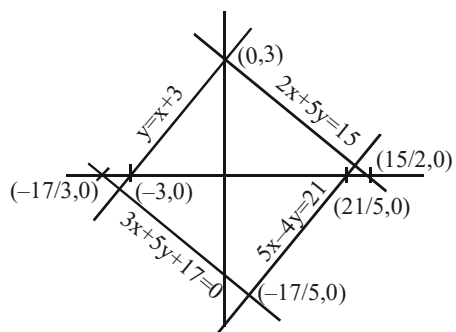
$$3x + 5y + 17 = 0 \text{ and } y = x + 3.$$

Then α can be :

- (A) -3 (B) 0 (C) 4 (D) 5

Ans. BC

Sol.



Plot the lines

$(\alpha, 0)$ is a point on x-axis so $\alpha \in \left(-3, \frac{21}{5}\right)$

- C01/2/024. If a line is such that it passes through the circumcentre of a triangle and sum of the algebraic perpendiculars on the line (from vertices) is equal to zero, then which of the following always holds :
- (A) line always passes through the centroid of the triangle
 (B) line always passes through the orthocentre of the triangle
 (C) line always passes through the incentre of the triangle
 (D) line always passes through one of the excentres of the triangle

Ans. AB

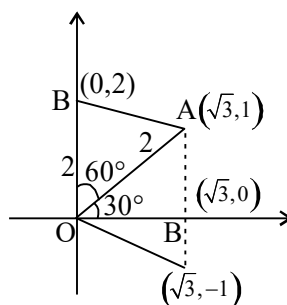
Sol. If sum algebraic length of perpendiculars is zero then line will pass through the centroid, as it also passes through circumcentre then orthocentre will also lie on the line because these 3 points are always collinear.

- C01/2/025. Let OAB is an equilateral triangle of side 2 units and one vertex at origin O. If OA is inclined at 30° to positive x-axis, then midpoint of AB has coordinates :

- (A) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right)$ (B) $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$ (C) $(\sqrt{3}, 0)$ (D) $(\sqrt{3}, -1)$

Ans. AC

Sol. M.P of AB is either $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right)$
 or $(\sqrt{3}, 0)$



- C01/2/026. For all values of ϕ , the lines represented by the equation $(2 \cos \phi + 3 \sin \phi)x + (3 \cos \phi - 5 \sin \phi)y - (5 \cos \phi - 2 \sin \phi) = 0$:
- (A) passes through the fixed point (a, b) such that $|2a - b| = 1$
 (B) passes through the fixed point $(2, 3)$
 (C) passes through a fixed point whose reflection with respect to line $x + y = \sqrt{2}$ is $(\sqrt{2} - 1, \sqrt{2} - 1)$
 (D) passes through the fixed point whose distance from the line $3x + 4y - 2 = 0$ is 1 unit

Ans. ACD

Sol. $(2 \cos \phi + 3 \sin \phi)x + (3 \cos \phi - 5 \sin \phi)y - (5 \cos \phi - 2 \sin \phi) = 0$
 $\cos \phi (2x + 3y - 5) + \sin \phi (3x - 5y + 2) = 0$

$$\begin{cases} 2x + 3y - 5 = 0 \\ 3x - 5y + 2 = 0 \end{cases}$$

passes through point of intersection of lines (1, 1)(B)

reflection of (1, 1) about $x + y = \sqrt{2}$

$$\frac{x-1}{1} = \frac{y-1}{1} = \frac{-2(1+1-\sqrt{2})}{2}$$

$$\frac{x-1}{1} = \sqrt{2} - 2, \frac{y-1}{1} = \sqrt{2} - 2$$

$$x = \sqrt{2} - 1, y = \sqrt{2} - 1$$

C01/2/027. Let $y = x + k$, ($k \neq 0$) is a chord AB of the curve $x^2 + xy - y^2 - x + 2y + 7 = 0$. If k_1 & k_2 are real values of k for which OA & OB (where O is origin) are equally inclined to x-axis, then which of the following is (are) true :

- (A) $k_1 + k_2 = 3$ (B) $k_1 + k_2 = 1$ (C) $k_1 k_2 = -14$ (D) $k_1 k_2 = 11$

Ans. AC

Sol. Homogenise the curve with the help of line

$$x^2 + xy - y^2 - x \left(\frac{y-x}{k} \right) + 2y \left(\frac{y-x}{k} \right) + 7 \left(\frac{y-x}{k} \right)^2 = 0$$

OA and OB are equally inclined to x-axis

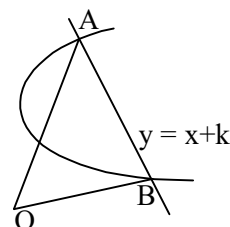
$$m_1 + m_2 = 0$$

$$1 - \frac{1}{k} - \frac{2}{k} - \frac{14}{k^2} = 0$$

$$k^2 - 3k - 14 = 0 \begin{cases} k_1 \\ k_2 \end{cases}$$

$$k_1 + k_2 = 3$$

$$k_1 k_2 = -14$$



C01/2/028. In a $\triangle ABC$ $A \equiv (\alpha, \beta)$, $B \equiv (1, 2)$, $C \equiv (2, 3)$ and point A lies on line $y = 2x + 3$, where $\alpha, \beta \in I$ and area of the triangle S is such that $2 \leq S < 3$. Then the coordinates of A will be :

- (A) $(-6, -9)$ (B) $(-7, -11)$ (C) $(2, 7)$ (D) $(3, 9)$

Ans. ABCD

Sol. $A(\alpha, \beta)$, $B(1, 2)$, $C(2, 3)$

$$\text{Eq. of BC : } (y - 2) = \frac{3-2}{2-1}(x - 1)$$

$$y - 2 = x - 1$$

$$y - x - 1 = 0$$

point A(h, 2h + 3)

$$\text{area of } \triangle ABC = \frac{1}{2} \times BC \times \text{height AD}$$

$$\text{Area} = \frac{1}{2} \times \sqrt{2} \times \left| \frac{2h+3-h-1}{\sqrt{2}} \right|$$

$$S = \frac{1}{2} |h+2|$$

$$\therefore 2 \leq S < 3$$

$$2 \leq \frac{1}{2} |h+2| < 3$$

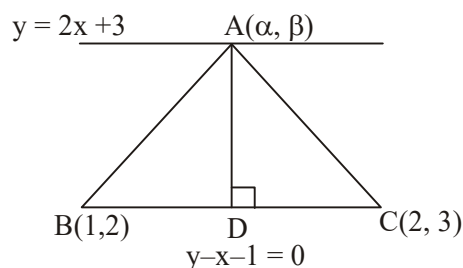
$$4 \leq |h+2| < 6$$

$$-4 \geq h+2 > -6 \text{ or } 4 \leq h+2 < 6$$

$$-8 < 8 \leq -6 \text{ or } 2 \leq h < 4$$

$$\therefore \text{Possible } h = -7, -6, 2, 3$$

$$\therefore A(-7, -11), (-6, -9), (2, 7), (3, 9)$$



C01/2/029. A line L is drawn from P(4, 3) to meet the lines $L_1 \equiv 3x + 4y + 5 = 0$ and $L_2 \equiv 3x + 4y + 15 = 0$ at points A and B respectively. From point A a line perpendicular to L is drawn meeting the line L_2 at A_1 . Similarly from point B, a line perpendicular to L is drawn meeting the line L_1 at B_1 . Thus a parallelogram AA_1BB_1 is formed. The equation of L so that area of parallelogram AA_1BB_1 is least :

(A) $x - 7y + 17 = 0$ (B) $7x + y - 31 = 0$ (C) $x - 7y - 31 = 0$ (D) $7x + y + 17 = 0$

Ans. AB

Sol. The given lines (L_1 and L_2) are parallel and distance between them (BC or AD) is $\frac{15-5}{5} = 2$ units

$$\text{Let } \angle BCA = \theta$$

$$\Rightarrow AB = BC \operatorname{cosec} \theta = 2 \operatorname{cosec} \theta \text{ and } AA_1 = AD \sec \theta = 2 \sec \theta$$

Now area of parallelogram AA_1BB_1 is

$$\Delta = AB \times AA_1 = 4 \sec \theta \cdot \operatorname{cosec} \theta$$

$$= \frac{8}{\sin 2\theta}$$

$$\text{Clearly, } \Delta \text{ is least for } \theta = \frac{\pi}{4}.$$

$$\text{Let slope of AB} = m$$

$$\text{then } 1 = \left| \frac{m + \frac{3}{4}}{1 - \frac{3m}{4}} \right|$$

$$\Rightarrow 4m + 3 = \pm(4 - 3m) \Rightarrow m = \frac{1}{7} \text{ or } -7$$

Hence the equation of 'L' is

$$x - 7y + 17 = 0 \text{ or } 7x + y - 31 = 0$$

C01/2/030. If algebraic sum of distances of a variable line from points A(3,0), B(0,3) and C(-3,-3) is zero, line is always passing through point (α, β) and (p,q) is centre of circumcircle of ΔABC , then :

- (A) $\alpha + \beta = 0$ (B) $\alpha + \beta = -1$ (C) $p + q = 0$ (D) $p + q = -1$

Ans. AD

Sol. As the sum of algebraic lengths of perpendiculars from C_1, C_2, C_4 and C_2, C_3, C_5 vanishes, so line $y = mx + c$ will pass through the centroid of $\Delta C_1 C_2 C_4$ and $\Delta C_2 C_3 C_5$, Now do yourself.

C02/2/031. The line $2x + y + 1 = 0$ intersects the circle $x^2 + y^2 = 9$ at two distinct points A and B and the line $3x + ky - 4 = 0$ intersects the circle $x^2 + y^2 - 2x + 4y + 1 = 0$ at C and D, then :

- (A) if A, B, C, D are concyclic, then k is -1.
(B) if A, B, C, D are concyclic, then k is 1.

(C) if a circle S passes through A, B, C, D, then radical centre of three circles is $\left(\frac{3}{5}, -\frac{11}{5}\right)$

(D) if a circle S passes through A, B, C, D, then radical centre of three circles is (-2,1)

Ans. AC

Sol. $S_1 : x^2 + y^2 - 9 = 0$
 $S_2 : x^2 + y^2 - 2x + 4y + 1 = 0$
 $S_1 - S_2 = 2x - 4y - 10 = 0$
 $x - 2y - 5 = 0$
 $(L_1, L_2, S_1 - S_2)$ are concurrent

$$\Rightarrow \begin{vmatrix} 1 & -2 & -5 \\ 2 & 1 & 1 \\ 3 & k & -4 \end{vmatrix} = 0$$

$$k = -1$$

$$\text{radical centre is } \left(\frac{3}{5}, -\frac{11}{5}\right)$$

